

Eli5: why are tech companies using nuclear energy to power ai?

Posted by u/Jncocontrol • 0 points • 11 comments

I saw some articles state Microsoft, Google and a few other were going to start using nuclear energy plants. But why? Doesn't AI work well with c++, that's from what I understand a very efficient computer language? And video game companies don't use it for example overwatch 2,, which would seem like the better candidate for their purpose

Comments

u/explainlikeimfive-ModTeam • 1 points • 2024-10-31 04:04 UTC

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u/kingjoey52a • 1 points • 2024-10-31 03:55 UTC

It's cheap, efficient, and safe. And it generates a lot of energy. We should be using nuclear power for everything but one Soviet plant blows up and everyone freaks out.

u/DartBen654 • 1 points • 2024-10-31 03:40 UTC

AI, as with all computing and processing, requires energy in the form of electricity. That electricity needs to be generated somehow. Tech companies generally prefer green (non-carbon emitting) energy. They also prefer their data centers to be powered 24/7/365. Outside of old school hydroelectric dams (think Hoover Dam), the only baseload firm generation in this country that doesn't emit carbon is a nuclear plant.

Batteries combined with solar or wind may get there eventually, but utility scale batteries are still rarer than nukes.

u/papersuite • 1 points • 2024-10-31 03:33 UTC

Being an efficient computing language and being energy efficient are two different things. To answer your question though let's me explain it like this

You have a small trailer. It can be towed by a small car. Now you have a larger trailer, and it can only be towed by a large truck or a semi-truck

You wouldn't use the small car to pull the large trailer because it is too heavy and would cause the car to break while not going anywhere.

Data centers are like the large trailer. They are often huge and require a powerful energy source. If a smaller energy source is used to power it, that source runs the risk of being overloaded, which can damage both the data center and the power source.

Nuclear is very very powerful and can produce the most power, just like the Semi-truck, so data centers use nuclear. The current wind and solar technology is sort of like the small car, it can pull small loads but not big ones without risking being overloaded and suffering damage.

There are other factors in play but this is the 10,000 ft view of the issue.

u/fiendishrabbit • 4 points • 2024-10-31 03:31 UTC

I don't think you realize the scale of these operations. There are 10 US states that use less electricity than Google.

u/sudoku7 • 1 points • 2024-10-31 03:29 UTC

"It's just that complicated" is probably the briefest answer. To start to answer why it is... Well you start going into the how the technology works which tends to be a "now let's talk about high school/college level computational complexity concepts."

One thing I think can be mentioned to maybe help clarify. While you can run a pre-trained model on a consumer GPU that will happily

run Overwatch 2, you can't train a model (ie, create that pre-trained model) on a consumer GPU currently and that may be part of your confusion.

u/ToolMeister • 1 points • 2024-10-31 03:28 UTC

Power comes from the grid, if a country has nuclear plants, everyone who has an electric device plugged into the grid would also be using nuclear power. This has nothing to do with programming languages

u/Aethonevg • 3 points • 2024-10-31 03:26 UTC

AI, or in this case LLM's use complex math algorithms to train the AI on gathered data to predict text generation. These math computations are extremely taxing and require the use of thousands of high power graphics cards in a server farm. These farms require a lot of power to keep running. Also efficiency of a programming language doesn't really translate in power efficiency. It's more so time.

u/ZimaGotchi • 1 points • 2024-10-31 03:25 UTC

AI is very energy hungry because its processes process up new processes to process, it's inherently exponential - and on top of that, everyone and their uncle wants to use it.

u/Sea-Dish-4766 • 11 points • 2024-10-31 03:25 UTC

The computation for AI is super energy intensive. Nuclear is the only high density power source that doesn't emit carbon.

u/[deleted] • 1 points • 2024-10-31 03:22 UTC

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